

Affine processes

A process whose characteristic function is exponentially affine in the current state:

$$E[e^{i\tau X_t} | X_0] = e^{a(s,t;\tau) + B(s,t;\tau) \cdot X_0}, \quad \forall a \in \mathbb{R}^d, \quad s \leq t$$

Usually we will assume time-homogeneous affine processes: $a(t;\tau) \rightarrow a(t;\tau), B(t;\tau) \rightarrow B(t;\tau)$

Motivating Example: Riccati ODEs derivation

Let's take a look at the simple scalar (d=1) square root process:

$$dX_t = (a+bX_t)dt + c\sqrt{X_t}dW_t$$

where $b \leq 0$ and the rest is positive. Objective is now to find the Laplace transform:

$$f(X_t, t) = E[e^{-\lambda X_t} | X_t]$$

Let $V_t = f(X_t, t)$. Then it should be clear that V_t is a martingale:

$$V_s = E[e^{-\lambda X_s} | X_s] = E[E[e^{-\lambda X_s} | X_t] | X_s] = E[V_t | X_s]$$

Now expand dV_t using Ito:

$$dV = \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial X}(a+bX) + \frac{1}{2} \frac{\partial^2 f}{\partial X^2} c^2 X \right) dt + \dots dW$$

The drift should be 0 because of Martingality. Now insert the affine ansatz: $f(x, t) = e^{a(t;\tau) + B(t;\tau)x}$ with the boundary conditions $a(0)=0, B(0)=\tau$ guaranteeing $f(x, T) = e^{-\lambda x}$:

$$t=T \rightarrow \frac{\partial f}{\partial t} = a'(t;\tau) + B'(t;\tau)x \rightarrow \frac{\partial f}{\partial t} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial X} \frac{\partial f}{\partial X} = \frac{\partial f}{\partial t}$$

$$\frac{\partial f}{\partial t} = -a'(s) + B'(s)x$$

$$\frac{\partial f}{\partial X} = B f$$

$$\frac{\partial^2 f}{\partial X^2} = B^2 f$$

$$-a'(s) - B'(s)x + B(a+bX) + \frac{1}{2} B^2 c^2 X = 0$$

Separation of variables: In order for this equation to hold for all x , we must require all terms containing x must be 0:

$$0 = -B'(s) + B(s)b + \frac{1}{2} B(s)^2 c^2 h_1 \quad / \quad B(0) = \tau$$

$$0 = -a'(s) + B(s)a + \frac{1}{2} B(s)^2 c^2 h_0 \quad / \quad a(0) = 0$$

These are Riccati ODEs that can be solved numerically.

We can also use the affine ansatz for other transforms, such as one that accounts for discounting:

$$f(X_t, t) = E[e^{-\int_t^T R(X_s) ds} | X_t], \quad R(x) = p_0 + p_1 x$$

Note that $f(X_t, t)$ is not a martingale, but $V_t = e^{-\int_t^T R(X_s) ds} f(X_t, t)$ is:

$$V_t = e^{-\int_t^T R(X_s) ds} E[e^{-\int_t^T R(X_s) ds} | X_t] = E[e^{-\int_t^T R(X_s) ds} | X_t]$$

Thus we can once again set the drift to 0, separate variables and find:

$$B'(s) = B(s)b + \frac{1}{2} B(s)^2 c^2 h_1 - p_1 \quad / \quad B(0) = \tau$$

$$a'(s) = a(s)b + \frac{1}{2} B(s)^2 c^2 h_0 - p_0 \quad / \quad a(0) = 0$$

We can extend to a multi-factor model:

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t$$

$$\mu(x) = K_0 + K_1 x$$

$$(\sigma(x)\sigma(x)^T) = H_{0ij} + H_{1ij} x_i x_j$$

And the Riccati equations become:

$$B'(s) = K_1^T B(s) + \frac{1}{2} B(s)^T H_1 B(s) - p_1, \quad B(0) = \tau$$

$$a'(s) = K_0^T B(s) + \frac{1}{2} B(s)^T H_0 B(s) - p_0, \quad a(0) = 0$$

Does not necessarily have explicit solution $\rightarrow$ solve numerically.

Possibly include general example in slides.

The conditional characteristic function of a random variable X is found by substituting $w=i\tau$ in the Laplace transform:

$$\phi_X = E[e^{i\tau X} | X_t]$$

The characteristic function is super useful, since we can recover the density function, through an inverse Fourier transform:

$$p(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ix\tau} \phi_X(\tau) d\tau$$

Or the cumulative distribution function using the Gil-Pelaez formula:

$$P(X \leq y) = \frac{1}{2} - \frac{1}{\pi} \int_0^{\infty} \frac{1}{\omega} \text{Im}[\phi_X(i\omega)] e^{i\omega y} d\omega$$

useful for pricing options, since $P(X \leq y) = E[\mathbb{1}_{X \leq y}]$

Non-uniqueness of Affine solution:

For any set of affine parameters (H, K, p) , we can find an equivalent set $(\hat{H}, \hat{K}, \hat{p})$ for different factors, that produces the same interest rates. Let $V_t = a + AX_t$ and choose:

$$\hat{p}_0 = p_0 - (A^{-1} p_1) \cdot a \quad \hat{p}_1 = A^{-1} p_1$$

Then we get: $p_0 + p_1 \cdot X_t = \hat{p}_0 + \hat{p}_1 \cdot X_t$. To fix this, we simply set $p_0 = 0$ and $p_1 = 1$, and instead tune only (H, K) .

What kind of conditions should we put on (H, K) to guarantee existence of a unique solution to SDE?

Essentially if the following condition holds:

$$\exists C \forall x, y: |\mu(x) - \mu(y)| + |\sigma(x) - \sigma(y)| \leq C(1 + |x|)$$

Along with the Lipschitz condition:

$$\exists D \forall x, y: |\mu(x) - \mu(y)| + |\sigma(x) - \sigma(y)| \leq D|x - y|$$

Then the SDE has a unique solution. Other condition in Duffie and Kan, but neither rules out that a unique solution doesn't exist.

Don't think about it too much.

What if we want jumps? Consider the SDE:

$$dX_t = (a+bX_t)dt + c\sqrt{X_t}dW_t + dJ_t$$

$$J_t = \sum_{i=1}^{N_t} Z_i \quad J_0 = 0$$

Z_i are i.i.d. with distribution function F_Z . The jump sizes (Z) are independent, but their stochastic intensity $\Lambda(x)$ is not:

$$\Lambda(x) = \lambda_0 + \lambda_1 x \quad (\text{if Poisson } \Lambda(x) = \lambda)$$

Let's try to find the Laplace Transform $\phi(X_t, t) = E[e^{-\lambda X_t} | X_t]$

Let $V_t = f(X_t, t)$ and apply Ito:

$$V_t = V_0 + \int_0^t \left(\frac{\partial f}{\partial s} + \frac{\partial f}{\partial X}(a+bX) + \frac{1}{2} \frac{\partial^2 f}{\partial X^2} c^2 X \right) ds + \int_0^t \frac{\partial f}{\partial X} c \sqrt{X} dW_s + \sum_{i=1}^{N_t} (f(X_t, t) - f(X_{t-}, t))$$

$$V_t = V_0 + M_t + A_t \quad \int_0^t \Lambda(X_s) ds = \Lambda(x) E[f(X_t, t) - f(X_{t-}, t)]$$

$$M_t = M_t^c + M_t^d = (\text{diffusion}) + (\text{Jumps} - \text{expected jumps})$$

$$\text{We are left with } A_t = \int_0^t (\lambda_0 f + \lambda_1 X f + \frac{1}{2} \frac{\partial^2 f}{\partial X^2} c^2 X) ds + \int_0^t \Lambda(X_s) ds$$

So for Martingality we must have:

$$\lambda_0 f + \lambda_1 X f + \frac{1}{2} \frac{\partial^2 f}{\partial X^2} c^2 X + \Lambda(x) E[f(X_t, t) - f(X_{t-}, t)] = 0$$

Use Affine ansatz: $f(x, t) = e^{a(t;\tau) + B(t;\tau)x}$:

$$\eta(x, t) = \Lambda(x) E[e^{a(t;\tau) + B(t;\tau)x} - e^{a(t;\tau) + B(t;\tau)x}] = \Lambda(x) E[e^{a(t;\tau) + B(t;\tau)x} - 1]$$

$$= (\lambda_0 + \lambda_1 x) f(x, t) (\theta(B(t;\tau)) - 1)$$

(Use $\theta(\omega) = E[e^{i\omega x}]$)

Final Riccati ODEs:

$$B'(s) = B(s)b + \frac{1}{2} B(s)^2 c^2 + \lambda_1 (\theta(B(s)) - 1) - p_1, \quad B(0) = \tau$$

$$a'(s) = a(s)b + \lambda_0 (\theta(B(s)) - 1) - p_0, \quad a(0) = 0$$

Multi-factor:

$$B'(s) = K_1^T B(s) + \frac{1}{2} B(s)^T H_1 B(s) + \lambda_1 (\theta(B(s)) - 1) - p_1$$

$$a'(s) = K_0^T B(s) + \frac{1}{2} B(s)^T H_0 B(s) + \lambda_0 (\theta(B(s)) - 1) - p_0$$

Pricing options is easy in the affine model. Consider the transform:

$$G_{a,b}(y; X_t, t, T, X) = E_t[e^{\int_t^T R(X_s) ds} e^{aX_T + bX_T}]$$

Pricing a Zero-Coupon put:

$$ZCP(t) = E_t[e^{\int_t^T R(X_s) ds} (K - P(T, T+; \tau))]$$

Note that $P(T, T+; \tau) = e^{a(\tau) + B(\tau)\tau}$, which gives us:

$$ZCP(t) = e^{a(t;\tau)} [G_{a,b}(t; (\log(K) - a(\tau)) + G_{a,b,0}(t; (\log(K) - a(\tau)))]$$

Note that: $e^{a(t;\tau) + B(t;\tau)\tau} \leq K \rightarrow B(t;\tau) \leq \log(K) - a(t;\tau)$

G -transform can be found using:

$$G_{a,b}(y; X_t, T, X) = \frac{\psi^X(a, X_t, t, T)}{\pi} \int_{-\infty}^{\infty} \tau \cdot \tau \cdot [\psi^X(a + i\tau b, X_t, t, T)] e^{-i\tau y} d\tau$$

Where $\psi^X(u; X_t, t, T) = E_t[e^{-\int_t^T R(X_s) ds} e^{uX_T}]$, which is solved through the Riccati ODEs.